

TECHNICAL WHITEPAPER

ASTRIA-CAT

Physics-Informed Edge AI Architecture for
Clear-Air Turbulence Prediction

Houssam Rharbi

Independent Systems Researcher

Version 2.0 — Architecture Specification

Edge AI / Atmospheric Physics / Hard-Real-Time Embedded Systems

github.com/fDg21G/ASTRIA-CAT

Abstract

ASTRIA-CAT V1 established feasibility: a one-dimensional convolutional neural network operating directly on raw differential pressure traces from a distributed MEMS sensor array, achieving sub-60 ms inference on embedded hardware. This configuration is a signal classifier. It carries no embedded representation of the governing fluid-dynamical mechanism — shear-driven stratified instability — and its output (a binary turbulence/no-turbulence likelihood) does not correspond to any quantity used in operational turbulence reporting.

ASTRIA-CAT V2 reformulates the problem from first principles. Raw pressure scalars are discarded in favor of dimensionless physical invariants — the pressure coefficient C_p and the gradient Richardson number R_i — which place the input space in direct correspondence with the Miles–Howard stability criterion governing Kelvin–Helmholtz instability (KHI) onset. The inference core transitions to a Spatiotemporal Graph Neural Network (ST-GNN): a fixed-topology sparse spatial operator over the sensor mesh, coupled to a dilated Temporal Convolutional Network (TCN) for multiscale transient capture. The output target is reformulated as a continuous regression of the Eddy Dissipation Rate, $EDR = \epsilon^{1/3}$, the quantity used in ICAO turbulence reporting.

The embedded execution path is rebuilt around hard-real-time guarantees: Quantization-Aware Training (QAT) to INT8, Ahead-of-Time (AOT) compilation via Apache TVM to a statically scheduled binary with zero dynamic memory allocation, and a worst-case execution time (WCET) bound suitable for certification-track avionics software. The result is a system that is physically interpretable, spatiotemporally consistent with the 4D structure of turbulence, and deterministic at the instruction level.

1 Atmospheric Physics & Nondimensionalization

1.1 The Failure Mode of Raw Pressure Input

A neural network trained on raw pressure values $p(t)$, expressed in pascals, learns a function that is implicitly conditioned on the operating point of the aircraft — altitude, true airspeed, and ambient density — because p itself is a dimensional quantity whose baseline and variance scale with all three. A model trained on data collected predominantly at FL350 will encounter a materially different input distribution at FL100, despite the underlying flow physics (a shear layer susceptible to KHI) being scale-invariant in its essential dynamics. This is a distributional generalization failure induced entirely by a poor choice of input representation, not by insufficient model capacity.

The correction is dimensional reduction at the input layer: every quantity entering the network must be expressed as a dimensionless group whose value is, to first order, independent of the

aircraft's instantaneous flight regime.

1.2 The Pressure Coefficient C_p

The first substitution replaces $p(t)$ with the pressure coefficient,

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V^2} \quad (1)$$

where p_∞ is the local freestream static pressure, ρ is local air density, and V is true airspeed. The denominator $\frac{1}{2}\rho V^2$ is the dynamic pressure q , itself directly available from the Air Data Inertial Reference Unit (ADIRU). C_p expresses each MEMS sensor reading as a perturbation *relative to the dynamic pressure scale of the current flight regime*, removing altitude and airspeed as confounding variables from the input distribution.

The high-frequency MEMS array therefore does not feed raw $p(t)$ to the network. It feeds $C_p(t)$, computed per sensor, per timestep, using a low-rate ADIRU stream for p_∞ , ρ , and V and a high-rate MEMS stream for p .

1.3 The Gradient Richardson Number R_i

C_p characterizes the perturbation field. It does not, by itself, characterize the *stability* of the background flow against which that perturbation evolves. For this, ASTRIA-CAT V2 introduces the gradient Richardson number,

$$R_i = \frac{N^2}{\left(\frac{\partial U}{\partial z}\right)^2} \quad (2)$$

where N is the Brunt–Väisälä frequency,

$$N^2 = \frac{g}{\theta} \frac{\partial \theta}{\partial z} \quad (3)$$

g is gravitational acceleration, θ is potential temperature, z is the vertical coordinate, and $\partial U / \partial z$ is the vertical gradient of horizontal wind speed (vertical wind shear).

R_i is the central diagnostic of stratified shear-flow stability. The Miles–Howard theorem establishes that a stratified shear flow is linearly unstable to Kelvin–Helmholtz perturbations only if

$$R_i < \frac{1}{4} \quad \text{somewhere in the flow.} \quad (4)$$

This is not a heuristic threshold tuned against a training set — it is a necessary condition derived from the linearized Taylor–Goldstein equation governing the perturbation structure $\hat{w}(z)$:

$$\frac{d^2 \hat{w}}{dz^2} + \left[\frac{N^2}{(U - c)^2} - \frac{1}{U - c} \frac{d^2 U}{dz^2} - k^2 \right] \hat{w} = 0 \quad (5)$$

where c is the perturbation phase speed and k is the horizontal wavenumber. $R_i < 1/4$ is the regime in which this equation admits growing solutions — the regime in which KHI billows can form.

Architectural consequence. R_i cannot be computed from pressure alone; it requires the temperature lapse rate and the wind shear profile, both of which are available from the aircraft’s existing ADIRU and static air temperature (SAT) sensors at low sample rate. ASTRIA-CAT V2 therefore defines a fusion boundary at the input layer: a low-rate physics channel (R_i , N^2 , vertical shear, computed once per ADIRU update cycle) is broadcast and concatenated with the high-rate sensor-mesh channel (C_p , sampled at MEMS rate). The network receives, at every timestep, both *what the perturbation looks like* (C_p) and *whether the background flow is dynamically capable of amplifying it* (R_i).

1.4 Input Tensor Formalization

For a sensor mesh of N_s nodes and a temporal window of length T , the per-timestep input to the spatiotemporal core is

$$\mathbf{x}_t \in \mathbb{R}^{N_s \times d}, \quad d = \{C_p^{(1)}, \dots, C_p^{(N_s)}, R_i, N^2, \partial U / \partial z\} \quad (6)$$

with R_i , N^2 , and $\partial U / \partial z$ broadcast identically across all N_s nodes (global flow-state context) and $C_p^{(i)}$ specific to sensor i (local perturbation field). This tensor is the unit of data ingested by the architecture described in Section 2.

2 Spatiotemporal Neural Architecture (Phase 2)

2.1 Rationale for a Spatiotemporal Graph Formulation

Clear-air turbulence is a four-dimensional phenomenon: three spatial dimensions plus time. A 1D-CNN operating on stacked sensor channels treats the channel axis as an unordered or arbitrarily-ordered set of features, discarding the physical adjacency structure of the sensor mesh on the airframe. KHI billows have a characteristic horizontal wavenumber k ; resolving any information about k requires comparing signals *across* sensors at known relative positions, not just within a single sensor’s time history.

ASTRIA-CAT V2 therefore represents the sensor mesh as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of N_s MEMS sensor nodes and $\mathcal{E}$ is defined by physical adjacency on the fuselage skin — each edge weighted by inverse inter-sensor distance. Spatial reasoning over $\mathcal{G}$, combined with temporal reasoning over the window T , constitutes the Spatiotemporal Graph Neural Network (ST-GNN) core.

2.2 Spatial Operator: Static Sparse Matrix Multiplication

A conventional GNN performs message passing with dynamically indexed neighbor lookups — a pattern that is fundamentally incompatible with the determinism requirements of Section 3. ASTRIA-CAT V2 resolves this by exploiting a property unique to this application: **the sensor topology is fixed at design time**. It is a physical property of the airframe and does not change at runtime.

The graph convolution therefore reduces to a single static sparse matrix multiplication. Given the normalized adjacency matrix $\hat{A} \in \mathbb{R}^{N_s \times N_s}$ (precomputed once, at compile time, from the known sensor geometry),

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2} \quad (7)$$

where A is the binary (or distance-weighted) adjacency matrix, I is the identity, and D is the degree matrix, the spatial propagation step is

$$H^{(l+1)} = \sigma \left(\hat{A} H^{(l)} W^{(l)} \right) \quad (8)$$

with $H^{(l)} \in \mathbb{R}^{N_s \times d_l}$ the node feature matrix at layer l , $W^{(l)}$ a learned dense weight matrix, and σ a nonlinearity. Because $\hat{A}$ is fixed and sparse, $\hat{A}H^{(l)}$ is not “graph message passing” at inference time — it is a constant-structure sparse–dense matmul with a compile-time-known sparsity pattern. This is the mechanism by which spatial gradient-like operators (the discrete analogue of $\nabla^2 C_p$) are realized without any dynamic graph traversal, dynamic indexing, or heap allocation.

2.3 Temporal Operator: Dilated Temporal Convolutional Network

Turbulence precursors span timescales from mesoscale gravity-wave modulation (minutes) to KHI billow rollover (sub-second to seconds) — several orders of magnitude. A fixed-kernel convolution cannot cover this range without either an impractically large kernel or an impractically deep stack. ASTRIA-CAT V2 uses a dilated causal TCN, where layer l has dilation factor

$$d_l = 2^l, \quad l = 0, 1, \dots, L - 1 \quad (9)$$

giving a receptive field of

$$R = 1 + 2(k - 1) \sum_{l=0}^{L-1} d_l = 1 + 2(k - 1)(2^L - 1) \quad (10)$$

for kernel size k . With $k = 3$ and $L = 8$, $R = 1021$ timesteps — covering both the high-frequency billow signature and the slower modulating envelope, with $O(L)$ layers rather than $O(R)$. Causality (each output depends only on present and past inputs) is preserved by construction, which is a precondition for any hard-real-time streaming inference path.

The full ST-GNN block at layer l is therefore

$$H_t^{(l+1)} = \sigma \left(\hat{A} \text{TCN}_{d_l} \left(H_{t-R_l:t}^{(l)} \right) W^{(l)} \right) \quad (11)$$

interleaving the fixed spatial operator with dilated temporal convolution at each block.

2.4 Output Reformulation: Eddy Dissipation Rate Regression

ASTRIA-CAT V1 produced a binary classification output (turbulence / no turbulence), a quantity with no standardized operational meaning. ASTRIA-CAT V2 regresses the Eddy Dissipation Rate,

$$EDR = \epsilon^{1/3} \quad (12)$$

where ϵ is the turbulent kinetic energy dissipation rate, the quantity reported in ICAO PIREP and DEVG turbulence metrics. ϵ is estimated from the inertial-subrange scaling of the second-order structure function of the C_p field,

$$D(r) = \left\langle [C_p(x + r) - C_p(x)]^2 \right\rangle \approx C (\epsilon r)^{2/3}, \quad r \text{ in the inertial subrange} \quad (13)$$

which provides the regression target during training. The network’s output head is therefore a single continuous scalar $\widehat{EDR}_t$ per timestep, rather than a softmax over discrete classes.

2.5 Physics-Consistency Loss

The total training loss combines a data term with a physics-consistency regularizer derived from Section 1:

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} \left(\widehat{EDR}_t, EDR_t \right) + \lambda \mathcal{L}_{\text{phys}} \quad (14)$$

$$\mathcal{L}_{\text{phys}} = \mathbb{E} \left[\widehat{EDR}_t \cdot \mathbb{I}(R_i > R_{i,\text{crit}}) \right], \quad R_{i,\text{crit}} = \frac{1}{4} \quad (15)$$

$\mathcal{L}_{\text{phys}}$ penalizes the network for predicting elevated dissipation rates in flow regimes where the Miles–Howard criterion indicates the background flow is linearly stable against KHI. This does not solve the Taylor–Goldstein equation at inference time; it constrains the learned mapping to remain consistent with the stability boundary it implies.

3 Hard-Real-Time Embedded Execution

3.1 From Average Latency to Worst-Case Execution Time

A reported average inference latency (e.g., “< 60 ms”) is not a meaningful figure for a system whose output may inform a flight-deck alert. The relevant quantity for certification-track avionics software is the **Worst-Case Execution Time (WCET)**: a provable upper bound on execution time across all possible inputs and control-flow paths, on the target processor, under the target memory configuration. ASTRIA-CAT V2’s embedded pipeline is designed so that WCET is not merely measured but *architecturally bounded* — every component is selected specifically because it admits static analysis.

3.2 Quantization-Aware Training (QAT, INT8)

Post-training quantization (PTQ) of a small-channel-count network produces unacceptable accuracy degradation: per-channel dynamic ranges in shallow networks are poorly represented by a single post-hoc scale factor. ASTRIA-CAT V2 instead applies Quantization-Aware Training, in which the forward pass simulates INT8 quantization,

$$\hat{x} = \text{clip} \left(\text{round} \left(\frac{x}{s} \right), -128, 127 \right) \cdot s \quad (16)$$

with scale s learned per channel during training, so that the network’s weights converge to values that are robust to the quantization noise they will experience at inference. The deployed model is therefore natively INT8 — not a degraded float model coerced into a smaller representation after the fact.

3.3 Ahead-of-Time Compilation via Apache TVM

The quantized model is exported to ONNX and compiled with Apache TVM in **Ahead-of-Time (AOT)** mode, rather than executed through a dynamic graph interpreter (e.g., the default ONNX Runtime execution provider). AOT compilation produces:

- A single statically scheduled executable function, with a fixed sequence of operations determined entirely at compile time.
- A fixed memory plan: every tensor’s size, lifetime, and address offset within a single pre-allocated arena are resolved before deployment.
- Zero dynamic (heap) allocation during inference — a hard requirement for memory-safety arguments in certification-track software.

The static-sparse-matmul formulation of the spatial operator (Section 2.2) is what makes this tractable: because $\hat{A}$ ’s sparsity pattern is fixed at compile time, the ST-GNN core compiles to the same class of statically schedulable kernel as the TCN — there is no data-dependent branching or variable-size intermediate tensor anywhere in the inference path.

3.4 Target Platform and Determinism Guarantees

Constraint	Implementation
Target processor	ARM Cortex-A / Cortex-R class embedded core
Numeric format	INT8 (QAT), fixed-point physics channel (R_i, N^2)
Memory allocation	Static arena, single allocation at initialization; zero <code>malloc</code> during inference loop
Control flow	No data-dependent branching; fixed instruction sequence per inference call
Spatial operator	Compile-time sparse matrix, fixed sparsity pattern
Temporal operator	Causal dilated TCN, fixed receptive field R
Compilation	Apache TVM, AOT mode, single exported binary artifact
Timing guarantee	Provable WCET bound via static analysis of the compiled schedule

3.5 Sensor Fusion Boundary

The embedded pipeline maintains two independent update cadences, corresponding to the two physical channels defined in Section 1.4: a high-rate MEMS path (C_p , sampled at sensor-array rate, feeding the ST-GNN spatial input) and a low-rate ADIRU/SAT path ($R_i, N^2, \partial U / \partial z$, computed once per ADIRU cycle and held constant across the high-rate window). Both paths write into fixed-offset locations within the static input tensor; no synchronization primitives, queues, or dynamic buffers are required, as both cadences are deterministic and known at compile time.